Quantitative Methods – Human Sciences

Backdoor adjustment (Answer key)

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Your name here

0.1 Preliminaries

In what follows, we will demonstrate the use and compare different methods to estimate causal effects in observational data, under the conditional ignorability of the treatment assumption. Let's denote units $i, \dots, N$, with treatment $D \in \{0, 1\}$, and potential outcomes Y_d. In particular, as is common in this literature, we want to estimate an Average Treatment Effect on the Treated (ATT), defined as: $E[\tau_i|D = 1] = E[Y_{1i}|D = 1] - E[Y_{0i}|D = 1]$.

We also observe a series of pre-treatment variables X_i. To estimate the ATT from observational data, in addition to SUTVA or consistency of the potential outcomes, we would need to assume the following:

- (Mean) Conditional independence of the control potential outcome and the treatment:

$$E[Y_{0i}] \perp\!\!\!\perp D_i|X_i$$

- Positivity or common support

$$0 < P(D|X) < 1$$

Under these conditions, we can estimate conditional treatment effects $E[Y_{1i} - Y_{0i}|X = x]$ for each X, and then recover the ATT using the adjustment formula:

$$\sum_{X_{D=1}} \mathbb{E}[\widehat{\tau}_i|X]P[X_{D=1}]$$

Where $X_{D=1}$ represent the covariate distribution among the treated.

0.2 Data

For this exercise, we will use the classical data in LaLonde (1986). This dataset has been widely used to evaluate the performance of different methods to estimate causal effects using observational data, since it has the unique feature of providing an experimental component that could be used as a benchmark, and an observational component that can be used to test the proposed method.

The experimental portion of the data comes from the National Supported Work Demonstration (NSW), a labor training program that randomly selected participants to treatment ($N_t = 185$) and control

($N_c = 260$) conditions. The observational data, in addition to the treated group from the experiment, includes information about additional subjects from the Panel Survey of Income Dynamics (PSID), who are potential observational controls ($N_{c,obs} = 2490$).

The data includes 10 raw covariates: age, education (years), race (white, black, hispanic), marriage status (married/not), high school diploma (yes/no), and both previous earnings and unemployment (during 1974 and 1975). The outcome of interest is earnings in 1978. To better illustrate the difference between methods, and following Hainmueller (2012), we will use an expanded version of this covariates, including their squares and every two-way interaction (65 covariates in total).

```
library("cobalt") # Package to assess covariate balance
```

```
 cobalt (Version 4.5.5, Build Date: 2024-04-02)
```

```
library("causalsens") # lalonde experimental vs lalonde observational
library("tidyverse") # Just for data manipulation
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
####################
### DATA LOADING ###
####################

# First, let's load the experimental dataset (lalonde.exp)
data(lalonde.exp, package="causalsens")
# Then also the observational dataset (lalonde.psid)
data(lalonde.psid, package="causalsens")

########################
### BASELINE BALANCE ###
########################
library("ebal") # Entropy balance (and matrixmaker function)
```

```
##
## ebal Package: Implements Entropy Balancing.

## See http://www.stanford.edu/~jhain/ for additional information.
```

```
# Let's expand the matrix of covariates
lalonde.exp.covs <- lalonde.exp |> select(-"treat",-"re78") |> matrixmaker()
lalonde.obs.covs <- lalonde.psid |> select(-"treat",-"re78") |> matrixmaker()
# Then list the covariates we will be using in the analysis
covs_raw <- lalonde.exp |> select(-re78, -treat) |> names()
covs <- names(lalonde.obs.covs)

# Create a single dataset with both obs and experimental data
# Just to plot balance in both
lalonde.mix <-
 rbind(lalonde.exp.covs |>
       mutate(treat = lalonde.exp$treat, wexp=1, wobs=ifelse(treat==1,1,0)),
      lalonde.obs.covs[lalonde.psid$treat==0,] |> # Observational controls
      mutate(treat=0, wexp=0, wobs=1)) # Weights for obs data
#names(lalonde.mix)

# Create "loveplot" to show covariate distribution in both samples
```

0.3 Balance

A first interesting inspection of the data is to assess the overall covariate balance in the different samples. As we can see in the first figure, in the experimental sample all the covariates are very well balanced among treated and controls. In the observational sample, on the other hand, treated and control subjects are very different, in particular with respect to their previous earning trajectories.

```
# The bal.table function takes the raw data and weights
# And outputs a table of standardized difference between treated and controls
# Love.plot takes a bal.table object as inputs
# And transform it into a plot to facilitate presentation

bal.tab(f.build("treat", covs), data=lalonde.mix,
      estimand = "ATT", m.threshold =.1,
      disp.var.ratio=FALSE, disp.means=TRUE,
      weights = c("wexp","wobs")) |> # Weigght for experimental and observational sample
 love.plot(abs=FALSE,
         sample.names =
           c("Combined","Experimental","Observational")) +
 labs(title= " ") + theme(legend.position = "top")
```

```
Warning: Large mean differences detected; you may not be using standardized
mean differences for continuous variables.
```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis.

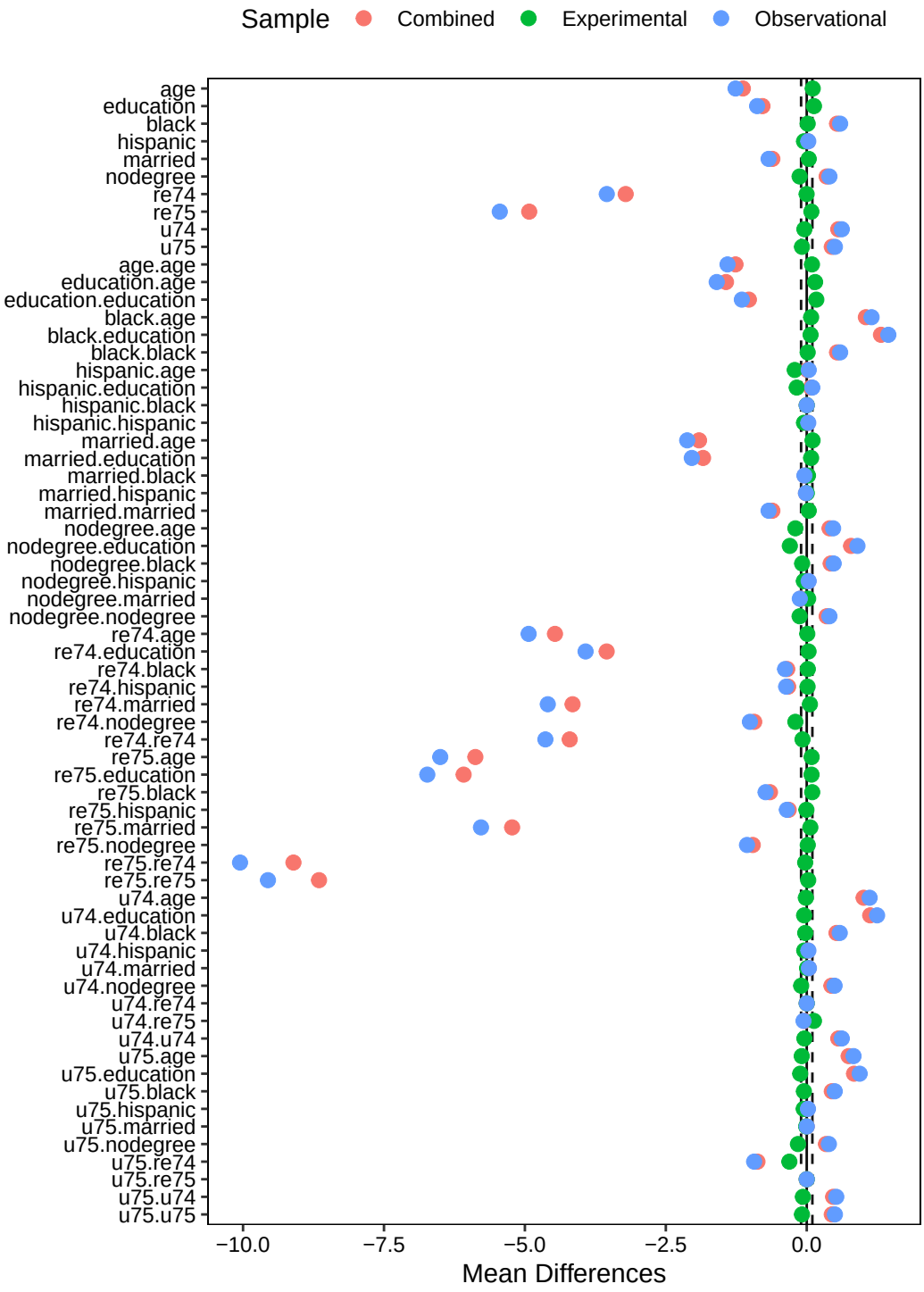


Figure 1: Covariate Balance by sample

0.4 Treatment effect

A second aspect that is worth noting are the estimated returns of the program, as we can see in the next figure. The experimental benchmark is around 1,600 and 1,800 USD, depending if we use a simple difference in means or the Lin estimator (a regression model with all raw covariates and their interactions with the treatment).

The observational difference in means, as expected from the huge covariate imbalance, is completely off: -15,205 USD. The multiple regression results with the observational data are more interesting, however. Although the point estimate is way below the point estimate from the experimental benchmark (suggesting only a 4USD return), the confidence interval resulting from the regression model overlaps with the benchmark returns. That's it, the regression model does a good job in accounting for the uncertainty associated to estimate the effect of the program, providing a wide and uninformative confidence interval.

```
######################
### BASELINE MODELS ###
######################
library("estimatr") # Lin estimator

# Bivariate regression on experimental data: 1,794 USD
exp_biv <- lm(re78 ~ treat, data=lalonde.exp)
# Lin estimator (treatment fully interacted with covs) on experimental data: 1,583 USD
exp_lin <- lm_lin(f.build("re78", "treat"), covariates = f.build("", covs_raw), data=lalonde.exp)
# Bivariate linear regression on observational data: -15,205 USD
obs_biv <- lm(re78 ~ treat, data=lalonde.psid)
# Multiple regression with controls on observational data: 4.159 USD
obs_reg <- lm(f.build("re78", c("treat", covs_raw)), data=lalonde.psid)
# Combine results (Store in a matrix for plotting)
level_order <- c("Experimental (diff-in-means)","Experimental (Lin estimator)",
          "Observational (Regression)","Observational (diff-in-means)")

# Compile the resulting coefficients into a single matrix
coeffs <-
 tibble(estimator =
        factor((c("Experimental (diff-in-means)",
          "Experimental (Lin estimator)",
          "Observational (Regression)",
          "Observational (diff-in-means)")), levels=level_order),
       coeff = c(exp_biv$coefficients[2],
             exp_lin$coefficients[2],
             obs_reg$coefficients[2],
             obs_biv$coefficients[2]),
       se = c(summary(exp_biv)$coefficients[2,2],
            exp_lin$std.error[2],
            summary(obs_reg)$coefficients[2,2],
            summary(obs_biv)$coefficients[2,2]))
```

```
# Create the annotated plot
ggplot(coeffs, aes(x=coeff, y=estimator)) +
 geom_point() +
 # Error bars (Confidence intervals)
 geom_errorbarh(aes(x=coeff, y=estimator,
                xmin=coeff-2*se, xmax=coeff+2*se),
            height=0) +
 geom_vline(xintercept = 0, linetype="dashed", color="red") +
 # Annotations (point estimates)
 annotate("text", x = -2500, y = 1, label = "1,794 USD") +
 annotate("text", x = -2500, y = 2, label = "1,583 USD") +
 annotate("text", x = -4000, y = 3, label = "4.159 USD") +
 annotate("text", x = -10000, y = 4, label = "-15,205 USD") +
 # Labels
 theme_classic() +
 labs(title = "Estimated return of the NSW training program",
     subtitle = "(Outcome: Earnings in 1978)",
     x = " ", y= " ")
```

Warning: `geom_errobarh()` was deprecated in ggplot2 4.0.0.
i Please use the `orientation` argument of `geom_errorbar()` instead.

`height` was translated to `width`.

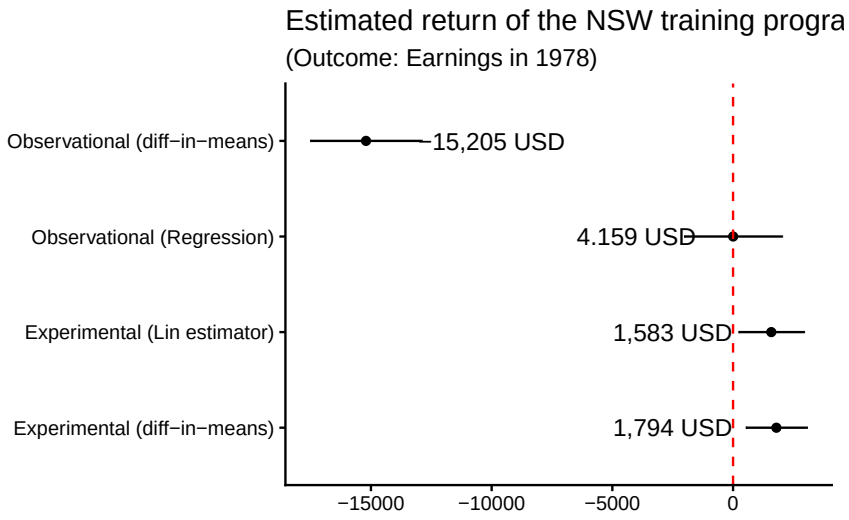


Figure 2: Estimated returns of the training program.

1 Overlap

Note: for this section, you may want to use the cobalt package in R. See reference here

Now it's your turn. First, we will assess overlap in the covariate distribution. Since it is not easy to do it directly with many covariates, you will need to estimate a *propensity score*. This is, a model for $P(D_i|X_i)$.

(a) Estimate a logistic regression with D as the outcome and the covs_raw as the predictors.

```
########################
### TABLE FORMATTING ###
########################
library("knitr")
library("kableExtra")
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
options(knitr.kable.NA = "")

# Format p-values for display
fmt_p <- function(p) ifelse(is.na(p), NA,
                    ifelse(p < 0.001, "<0.001", sprintf("%.3f", p)))

# Format numbers to 4 significant digits (keeps very small and very large
# coefficients readable, e.g. effects of earnings measured in dollars)
fmt_num <- function(x) ifelse(is.na(x), NA,
                      formatC(x, digits = 4, format = "g", big.mark = ","))

# Print a data frame / matrix as a LaTeX table for the PDF.
# - wide = TRUE shrinks tables that are wider than the page
# - long = TRUE allows tables to break across pages
kbl_pdf <- function(x, caption = NULL, digits = 3, wide = FALSE, long = FALSE, ...) {
 # Data frames without row names (e.g. coefficient tables): first column
 # left-aligned, all others right-aligned (also for pre-formatted numbers)
 align <- NULL
 if (is.data.frame(x) && .row_names_info(x) < 0) {
  align <- c("l", rep("r", ncol(x) - 1))
 }
 out <- kbl(x, format = "latex", booktabs = TRUE, digits = digits, align = align,
         caption = caption, label = NA, longtable = long, linesep = "",
         format.args = list(big.mark = ","), ...)
 if (long) {
  kable_styling(out, latex_options = c("repeat_header"), font_size = 9)
 } else if (wide) {
```

```
  kable_styling(out, latex_options = c("hold_position", "scale_down"))
 } else {
  kable_styling(out, latex_options = c("hold_position"))
 }
}

# Coefficient table from a fitted glm
coef_table <- function(model) {
 broom::tidy(model) |>
  mutate(across(c(estimate, std.error, statistic), fmt_num),
         p.value = fmt_p(p.value)) |>
  rename(Term = term, Estimate = estimate, `Std. Error` = std.error,
         `z value` = statistic, `p-value` = p.value)
}

# Propensity score model with the 10 raw covariates
# (observational data: NSW treated + PSID controls)
ps_raw <- glm(f.build("treat", covs_raw),
              data = lalonde.psid,
              family = binomial(link = "logit"))
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
coef_table(ps_raw) |>
 kbl_pdf(caption = "Propensity score model (raw covariates)")
```

Table 1: Propensity score model (raw covariates)

Term	Estimate	Std. Error	z value	p-value
(Intercept)	0.3715	1.282	0.2898	0.772
age	-0.1053	0.01775	-5.931	<0.001
education	-0.01316	0.076	-0.1732	0.862
black	2.611	0.3608	7.238	<0.001
hispanic	2.164	0.5928	3.65	<0.001
married	-1.573	0.2833	-5.553	<0.001
nodegree	0.6404	0.3764	1.701	0.089
re74	2.233e-05	3.109e-05	0.718	0.473
re75	-0.0002758	4.895e-05	-5.635	<0.001
u74	3.233	0.4926	6.563	<0.001
u75	-1.372	0.4599	-2.983	0.003

(b) Estimate an alternative logistic regression, with D as th outcome and covs as the predictors.

```
# Observational dataset with the expanded covariates (65 terms),
# plus treatment and outcome
lalonde.obs <- lalonde.obs.covs |>
  mutate(treat = lalonde.psid$treat,
         re78  = lalonde.psid$re78)

# Propensity score model with the expanded covariates.
# Note: some expanded terms are redundant (e.g., black.black = black) or
# constant (e.g., hispanic.black = 0), so their coefficients are NA, and
# R may warn that fitted probabilities of 0 or 1 occurred.
ps_full <- glm(f.build("treat", covs),
               data = lalonde.obs,
               family = binomial(link = "logit"),
               control = glm.control(maxit = 100))
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
# NA rows correspond to the redundant/constant terms dropped from the model
coef_table(ps_full) |>
  kbl_pdf(caption = "Propensity score model (expanded covariates)", long = TRUE)
```

Table 2: Propensity score model (expanded covariates)

Term	Estimate	Std. Error	z value	p-value
(Intercept)	1.752e+15	9.223e+07	1.9e+07	<0.001
age	8.268e+13	1.633e+06	5.065e+07	<0.001
education	-1.961e+14	1.139e+07	-1.721e+07	<0.001
black	-1.955e+15	2.93e+07	-6.671e+07	<0.001
hispanic	2.332e+15	5.878e+07	3.967e+07	<0.001
married	-1.8e+15	3.201e+07	-5.623e+07	<0.001
nodegree	-7.954e+14	5.859e+07	-1.357e+07	<0.001
re74	1.46e+11	1,948	7.494e+07	<0.001
re75	-1.519e+11	1,845	-8.235e+07	<0.001
u74	1.023e+15	5.896e+07	1.735e+07	<0.001
u75	1.825e+13	5.55e+07	3.288e+05	<0.001
age.age	-1.845e+12	1.618e+04	-1.14e+08	<0.001
education.age	3.149e+11	8.251e+04	3.817e+06	<0.001
education.education	2.622e+11	3.661e+05	7.162e+05	<0.001
black.age	3.032e+13	3.844e+05	7.886e+07	<0.001
black.education	1.252e+14	1.958e+06	6.395e+07	<0.001
black.black				
hispanic.age	-4.135e+13	8.181e+05	-5.055e+07	<0.001
hispanic.education	-6.821e+13	3.644e+06	-1.872e+07	<0.001
hispanic.black				
hispanic.hispanic				
married.age	-4.598e+12	4.546e+05	-1.011e+07	<0.001
married.education	8.181e+13	2.031e+06	4.028e+07	<0.001
married.black	-6.331e+13	9.108e+06	-6.95e+06	<0.001
married.hispanic	3.694e+14	2.44e+07	1.514e+07	<0.001
married.married				
nodegree.age	-1.818e+13	4.932e+05	-3.686e+07	<0.001

Table 2: Propensity score model (expanded covariates) *(continued)*

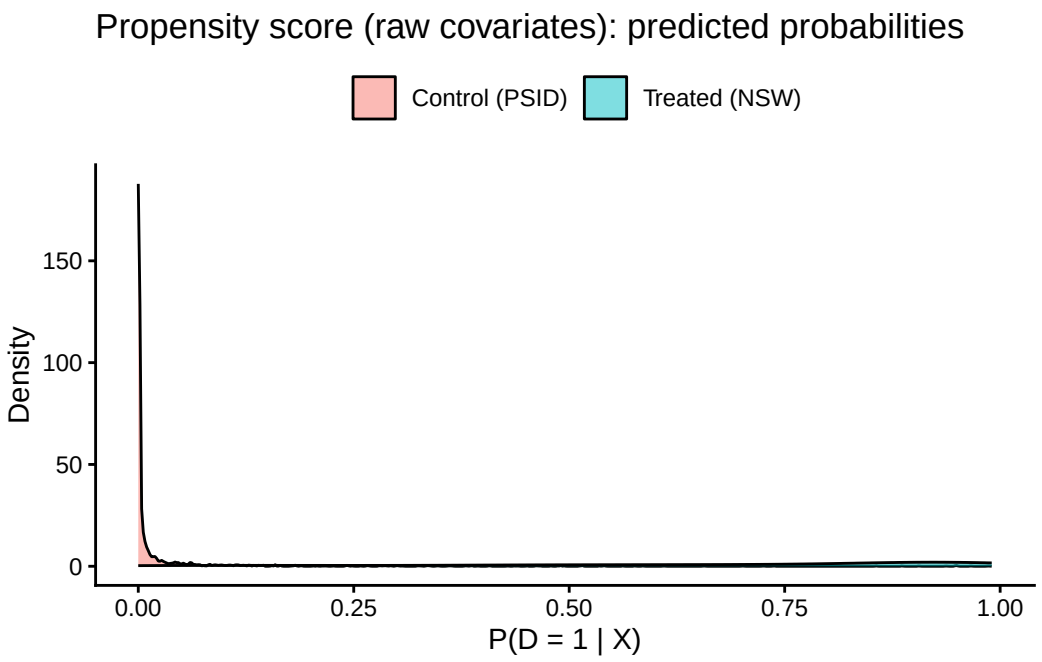
Term	Estimate	Std. Error	z value	p-value
nodegree.education	6.789e+13	4.869e+06	1.394e+07	<0.001
nodegree.black	3.049e+14	1.041e+07	2.928e+07	<0.001
nodegree.hispanic	-1.781e+13	2.444e+07	-7.287e+05	<0.001
nodegree.married	1.807e+14	1.223e+07	1.478e+07	<0.001
nodegree.nodegree				
re74.age	-1.218e+09	25.51	-4.776e+07	<0.001
re74.education	-9.095e+09	118.2	-7.695e+07	<0.001
re74.black	5.476e+09	633.5	8.644e+06	<0.001
re74.hispanic	-7.404e+10	1,188	-6.23e+07	<0.001
re74.married	-3.001e+10	698.1	-4.299e+07	<0.001
re74.nodegree	-5.745e+10	771.6	-7.446e+07	<0.001
re74.re74	1.066e+06	0.01262	8.45e+07	<0.001
re75.age	1.458e+09	24.69	5.907e+07	<0.001
re75.education	-1.778e+08	112.4	-1.581e+06	<0.001
re75.black	-1.08e+10	627.9	-1.72e+07	<0.001
re75.hispanic	2.069e+10	908.9	2.276e+07	<0.001
re75.married	1.087e+11	695.7	1.563e+08	<0.001
re75.nodegree	4.902e+10	723	6.78e+07	<0.001
re75.re74	-5.96e+05	0.01832	-3.253e+07	<0.001
re75.re75	-1.44e+05	0.01161	-1.24e+07	<0.001
u74.age	2.171e+13	7.409e+05	2.93e+07	<0.001
u74.education	-7.285e+13	3.619e+06	-2.013e+07	<0.001
u74.black	-6.164e+14	1.712e+07	-3.601e+07	<0.001
u74.hispanic	-1.095e+15	3.626e+07	-3.02e+07	<0.001
u74.married	-5.106e+14	1.756e+07	-2.907e+07	<0.001
u74.nodegree	4.245e+14	2.192e+07	1.937e+07	<0.001
u74.re74				
u74.re75	-6.17e+10	867.2	-7.114e+07	<0.001
u74.u74				
u75.age	-3.036e+13	7.052e+05	-4.305e+07	<0.001
u75.education	1.208e+13	3.421e+06	3.532e+06	<0.001
u75.black	-1.038e+14	1.644e+07	-6.313e+06	<0.001
u75.hispanic	-1.484e+14	3.169e+07	-4.683e+06	<0.001
u75.married	6.285e+14	1.723e+07	3.647e+07	<0.001
u75.nodegree	1.517e+14	2.093e+07	7.248e+06	<0.001
u75.re74	-8.916e+09	843.3	-1.057e+07	<0.001
u75.re75				
u75.u74	1.341e+14	1.852e+07	7.243e+06	<0.001
u75.u75				

(c) Now, from the model in (a), get both the predicted probabilities ($\in [0, 1]$) and the logits ($\in [-\infty, +\infty]$). Then create a density plot overlying the distribution of the predicted probabilities for treated and the *observational* control units. Do the same with the logits.

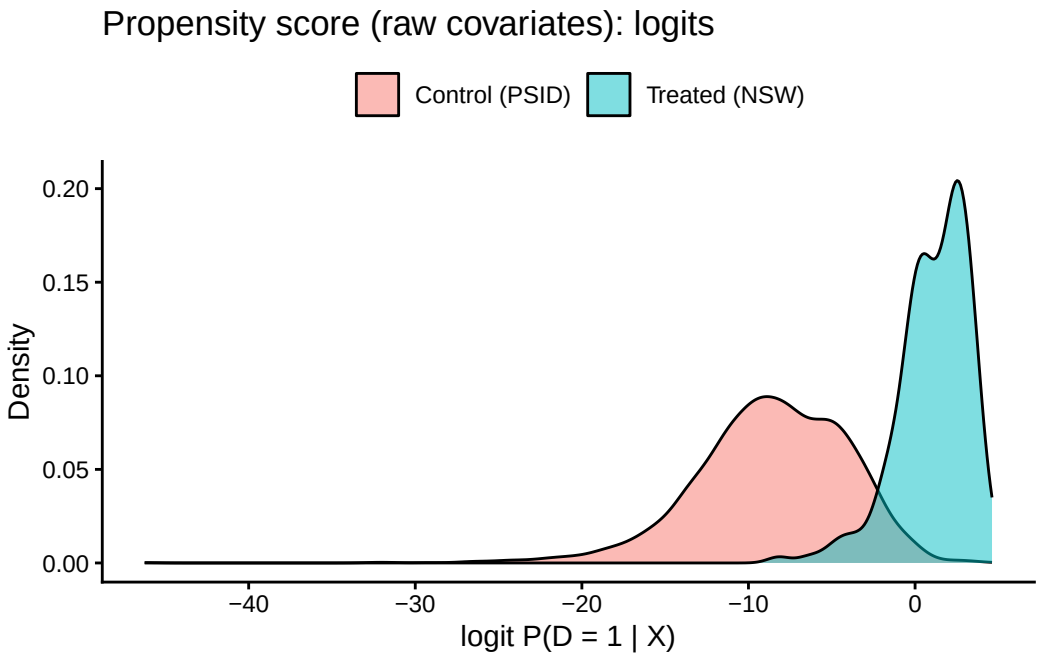
```
# Predicted probabilities and logits (linear predictor) from model (a)
ps_raw_df <- tibble(
  treat = factor(lalonde.psid$treat, levels = c(0, 1),
                 labels = c("Control (PSID)", "Treated (NSW)")),
  prob  = predict(ps_raw, type = "response"),
  logit = predict(ps_raw, type = "link")
)

# Density of predicted probabilities
```

```
ggplot(ps_raw_df, aes(x = prob, fill = treat)) +
  geom_density(alpha = 0.5) +
  theme_classic() +
  labs(title = "Propensity score (raw covariates): predicted probabilities",
       x = "P(D = 1 | X)", y = "Density", fill = " ") +
  theme(legend.position = "top")
```



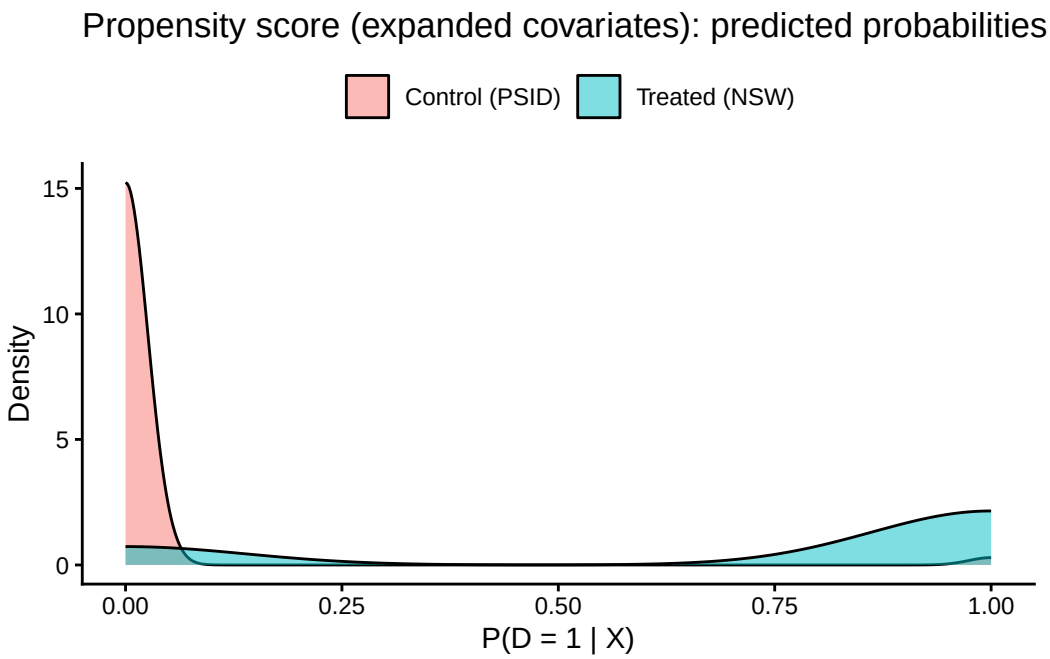

```
# Density of logits
ggplot(ps_raw_df, aes(x = logit, fill = treat)) +
  geom_density(alpha = 0.5) +
  theme_classic() +
  labs(title = "Propensity score (raw covariates): logits",
       x = "logit P(D = 1 | X)", y = "Density", fill = " ") +
  theme(legend.position = "top")
```



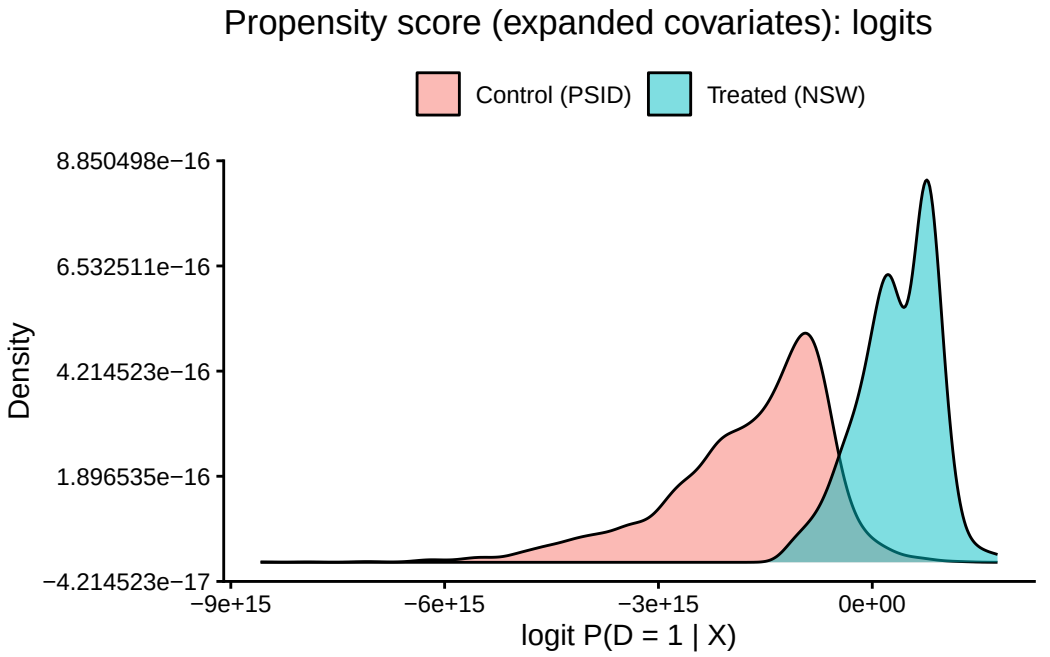
(d) Now, from the model in (b), get both the predicted probabilities ($\in [0, 1]$) and the logits ($\in [-\infty, +\infty]$). Then create a density plot overlying the distribution of the predicted probabilities for treated and the *observational* control units. Do the same with the logits.

```
# Predicted probabilities and logits from model (b)
# (taken directly from the fitted model, which avoids the
#  rank-deficiency warning that predict() would give)
ps_full_df <- tibble(
  treat = factor(lalonde.obs$treat, levels = c(0, 1),
                 labels = c("Control (PSID)", "Treated (NSW)")),
  prob  = ps_full$fitted.values,
  logit = ps_full$linear.predictors
)

# Density of predicted probabilities
ggplot(ps_full_df, aes(x = prob, fill = treat)) +
  geom_density(alpha = 0.5) +
  theme_classic() +
  labs(title = "Propensity score (expanded covariates): predicted probabilities",
       x = "P(D = 1 | X)", y = "Density", fill = " ") +
  theme(legend.position = "top")
```




```
# Density of logits
ggplot(ps_full_df, aes(x = logit, fill = treat)) +
  geom_density(alpha = 0.5) +
  theme_classic() +
  labs(title = "Propensity score (expanded covariates): logits",
       x = "logit P(D = 1 | X)", y = "Density", fill = " ") +
  theme(legend.position = "top")
```



(e) What can you conclude from the previous exercise?

2 Matching

For this section, you are asked to use the MatchIt package in R to estimate the Average Treatment Effect on the Treated using a matching approach. You can find documentation and tutorials here.

(a) Select one distance measure and one matching method. Explain how they work, and justify why do you think these are the most sensible choices.

(b) Perform the matching procedure using the matchit routine. Show your results (i.e., print the matching object)

```
library("MatchIt")
```

Attaching package: 'MatchIt'

The following object is masked from 'package:cobalt':

lalonde

```
# Distance: propensity score from a logistic regression on the raw covariates
# Method:   1:1 nearest neighbor matching (without replacement)
set.seed(1234)
m.out <- matchit(f.build("treat", covs_raw),
                 data     = lalonde.psid,
                 method   = "nearest",
                 distance = "glm",
                 estimand = "ATT",
                 ratio    = 1,
                 replace  = FALSE)
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
m.out
```

A `matchit` object
- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
    - estimated with logistic regression
- number of obs.: 2675 (original), 370 (matched)
- target estimand: ATT
- covariates: age, education, black, hispanic, married, nodegree, re74, re75, u74, u75

(c) Assess the balance obtained: i. using summary on your matching object; ii. plot the distribution of the propensity scores using the type = "jitter" option; iii. plot the balance of covariates using the type = "density" option; iv. plot the balance of covariates calling the function on the summary of your matching object. Interpret all outputs.

```
# i. Balance summary (before and after matching)
m.sum <- summary(m.out)

m.sum$sum.all |>
 kbl_pdf(caption = "Covariate balance: all data", wide = TRUE)
```

Table 3: Covariate balance: all data

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean	eCDF Max	Std. Pair Dist.
distance	0.685	0.023	2.215	11.012	0.485	0.887	
age	25.816	34.851	-1.263	0.470	0.232	0.377	
education	10.346	12.117	-0.881	0.425	0.109	0.403	
black	0.843	0.251	1.630		0.593	0.593	
hispanic	0.059	0.033	0.114		0.027	0.027	
married	0.189	0.866	-1.729		0.677	0.677	
nodegree	0.708	0.305	0.886		0.403	0.403	
re74	2,095.574	19,428.746	-3.547	0.133	0.468	0.729	
re75	1,532.056	19,063.338	-5.446	0.056	0.469	0.774	
u74	0.708	0.086	1.368		0.622	0.622	
u75	0.600	0.100	1.021		0.500	0.500	

```
m.sum$sum.matched |>
 kbl_pdf(caption = "Covariate balance: matched data", wide = TRUE)
```

Table 4: Covariate balance: matched data

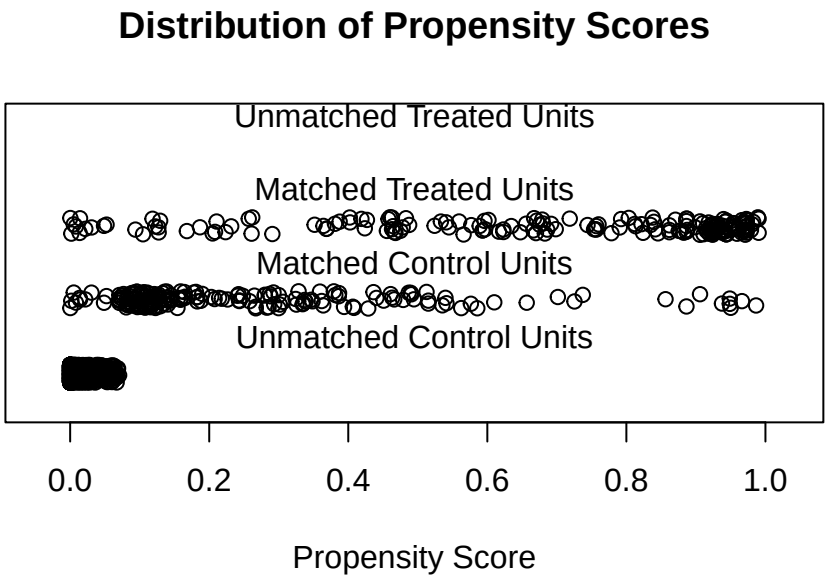
	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean	eCDF Max	Std. Pair Dist.
distance	0.685	0.255	1.438	1.805	0.048	0.622	1.438
age	25.816	29.573	-0.525	0.452	0.096	0.135	1.349
education	10.346	10.546	-0.099	0.473	0.043	0.097	1.282
black	0.843	0.703	0.387		0.141	0.141	0.892
hispanic	0.059	0.065	-0.023		0.005	0.005	0.526
married	0.189	0.508	-0.814		0.319	0.319	1.090
nodegree	0.708	0.611	0.214		0.097	0.097	0.904
re74	2,095.574	4,791.689	-0.552	0.697	0.079	0.330	1.016
re75	1,532.056	3,331.858	-0.559	0.733	0.065	0.314	0.980
u74	0.708	0.405	0.666		0.303	0.303	1.094
u75	0.600	0.395	0.419		0.205	0.205	1.103

```
m.sum$nn |>
 kbl_pdf(caption = "Sample sizes", digits = 0)
```

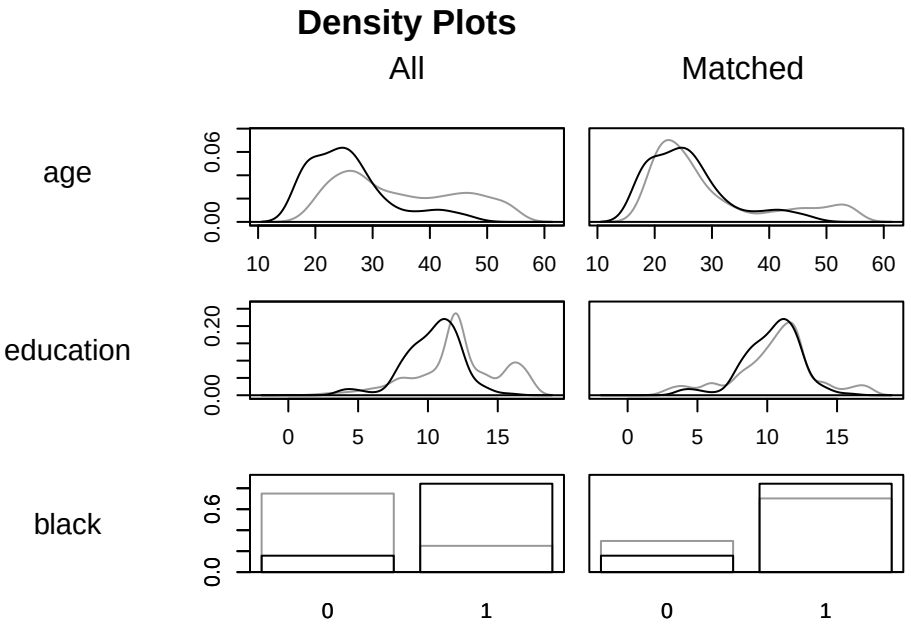
Table 5: Sample sizes

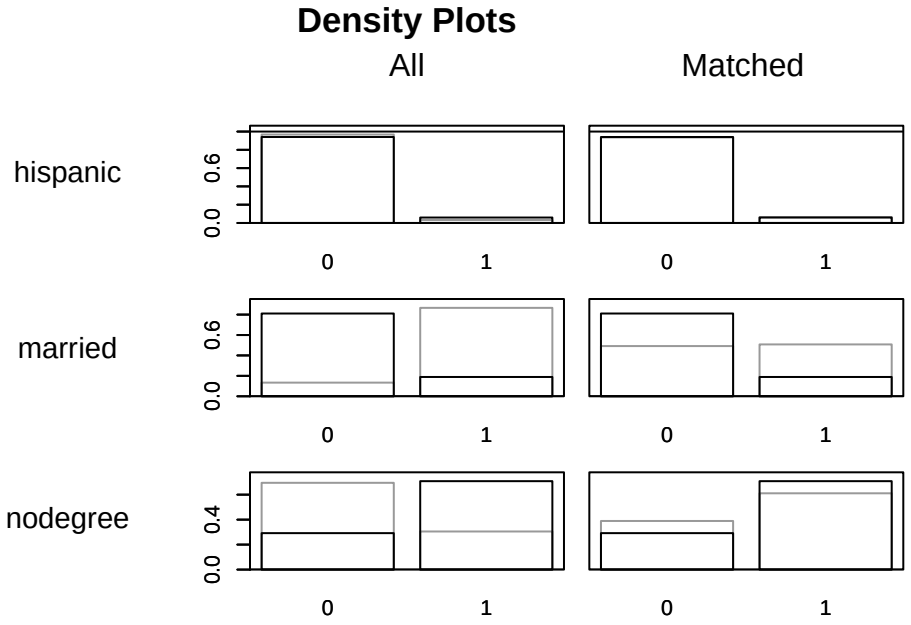
	Control	Treated
All (ESS)	2,490	185
All	2,490	185
Matched (ESS)	185	185
Matched	185	185
Unmatched	2,305	0
Discarded	0	0

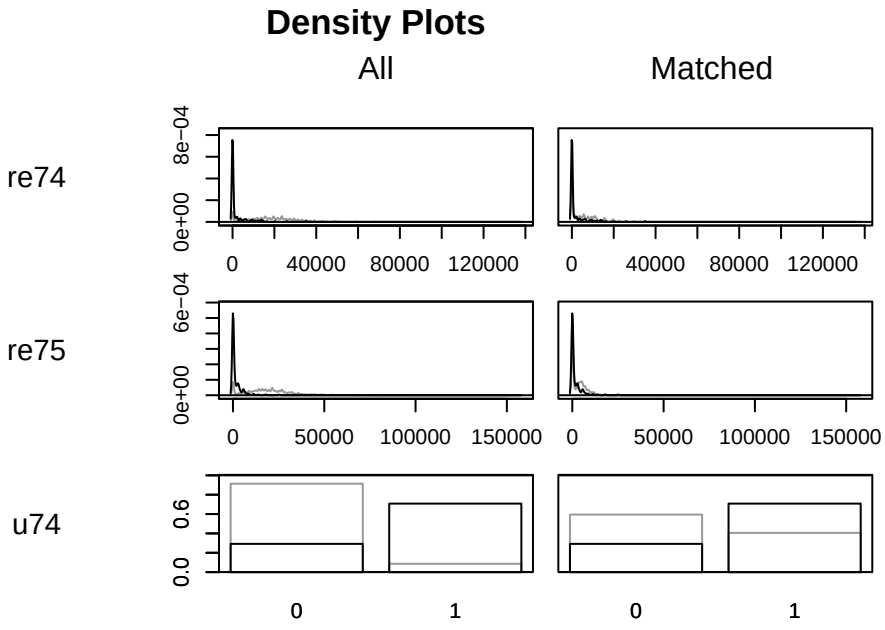
```
# ii. Distribution of propensity scores (matched vs. unmatched units)
plot(m.out, type = "jitter", interactive = FALSE)
```

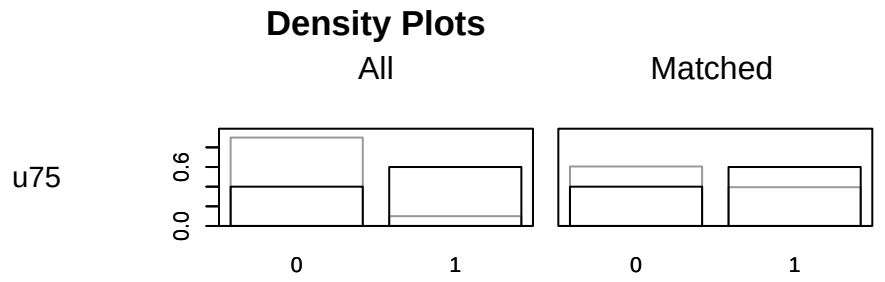



```
# iii. Covariate distributions before and after matching
plot(m.out, type = "density", interactive = FALSE)
```

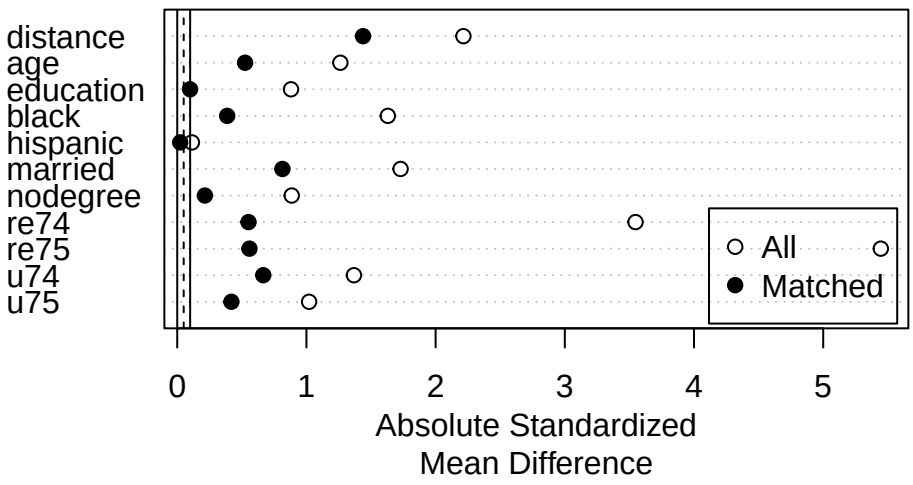

Density Plots
All
Matched
age
education
black
0.00
0.06
0.20
0.6
0.0
10
20
30
40
50
60
0
5
10
15
0
1


Density Plots
All
Matched
hispanic
married
nodegree
0.0
0.6
0.4
0
1






```
# iv. Love plot of standardized mean differences (from the summary object)
plot(summary(m.out))
```



(d) Estimate the treatment effect, and interpret the results.

```
library("marginaleffects")

# Table of estimated effects from avg_comparisons()
ate_table <- function(cmp) {
  as.data.frame(cmp) |>
    transmute(Term = term, Contrast = contrast, Estimate = estimate,
              `Std. Error` = std.error, z = statistic,
              `p-value` = fmt_p(p.value),
              `CI lower` = conf.low, `CI upper` = conf.high)
}

# Extract the matched data (includes matching weights and pair membership)
m.data <- match.data(m.out)

# Simple difference in means in the matched sample,
# with standard errors clustered by matched pair
fit_m_dim <- lm(re78 ~ treat, data = m.data, weights = weights)
lmtest::coeftest(fit_m_dim, vcov. = sandwich::vcovCL, cluster = ~subclass) |>
  coef_table() |>
  rename(`t value` = `z value`) |>
  kbl_pdf(caption = "Matching: difference in means (pair-clustered SEs)")
```

```
# Regression adjustment in the matched sample (treatment interacted with
# covariates), averaging the effect over the treated units (ATT),
# with standard errors clustered by matched pair
```

Table 6: Matching: difference in means (pair-clustered SEs)

Term	Estimate	Std. Error	t value	p-value
(Intercept)	7,326	821.8	8.915	<0.001
treat	-977.1	1,023	-0.9553	0.340

```
fit_m <- lm(as.formula(paste0("re78 ~ treat * (", paste(covs_raw, collapse = " + "), ")")),
        data = m.data, weights = weights)

avg_comparisons(fit_m,
            variables = "treat",
            vcov      = ~subclass,
            newdata   = subset(m.data, treat == 1),
            wts       = "weights") |>
 ate_table() |>
 kbl_pdf(caption = "Matching: regression-adjusted ATT (pair-clustered SEs)", digits = 2)
```

Table 7: Matching: regression-adjusted ATT (pair-clustered SEs)

Term	Contrast	Estimate	Std. Error	z	p-value	CI lower	CI upper
treat	1 - 0	2,128.79	1,182.56	1.8	0.072	-188.99	4,446.57

2.1 Extensions

(e) Repeat b) to d) using an alternative propensity score specification, and a different matching method.

(f) How do results compare with the previous? Assuming you *cannot* see the experimental benchmark, what estimate would you prefer and why?

3 Weighting

For this section, you are asked to use the WeightIt package in R to estimate the Average Treatment Effect on the Treated using a weighting approach. You can find documentation and tutorials here.

(a) Select two balancing methods, one based on the direct estimation of the propensity score (balance *in expectation*), and one directly targeting balance *in sample*. Explain how they work, and justify why do you think these are the most sensible choices.

```
library("WeightIt")

# Balancing methods used below:
#  - "glm":  inverse probability weights from a logistic propensity score
#          (balance in expectation)
#  - "ebal": entropy balancing, which solves for weights that exactly match
#          the covariate means of the treated group (balance in sample)
# Both use the raw covariates and target the ATT.
f_w <- f.build("treat", covs_raw)
f_w
```

```
treat ~ age + education + black + hispanic + married + nodegree +
  re74 + re75 + u74 + u75
<environment: 0x12c3c9148>
```

(b) Perform the weighting procedure using the weightit routine. Show your results (i.e., print the output object)

```
# Propensity score weighting (logistic regression)
W.glm <- weightit(f_w,
             data     = lalonde.psid,
             method   = "glm",
             estimand = "ATT")
```

```
Warning: Propensity scores numerically equal to 0 or 1 were estimated,
indicating perfect separation and infinite parameter estimates. These
may yield problems with inference. Consider trying a different `link`.
See `help("method_glm", package = "WeightIt")` for details.
```

```
Warning: Some extreme weights were generated. Examine them with `summary()` and
maybe trim them with `trim()`.
```

```
W.glm
```

```
A weightit object
 - method: "glm" (propensity score weighting with GLM)
 - number of obs.: 2675
 - sampling weights: none
 - treatment: 2-category
 - estimand: ATT (focal: 1)
 - covariates: age, education, black, hispanic, married, nodegree, re74, re75, u74, u75
```

```
# Entropy balancing
W.ebal <- weightit(f_w,
                   data     = lalonde.psid,
                   method   = "ebal",
                   estimand = "ATT")
```

Warning: Some extreme weights were generated. Examine them with `summary()` and maybe trim them with `trim()`.

```
W.ebal
```

A weightit object
 - method: "ebal" (entropy balancing)
 - number of obs.: 2675
 - sampling weights: none
 - treatment: 2-category
 - estimand: ATT (focal: 1)
 - covariates: age, education, black, hispanic, married, nodegree, re74, re75, u74, u75

(c) Assess the balance obtained: i. using summary on your weighting object; ii. using bal.tab on your weighting object; iii. creating a plot showing mean balance for all covariates, using the cobalt::love.plot() function. Interpret all outputs.

```
# Helper: tables from summary() of a weightit object
weights_table <- function(sw) {
 rng <- sapply(sw$weight.range, range)
 out <- rbind(`Min. weight`       = rng[1, ],
              `Max. weight`       = rng[2, ],
              `Coef. of variation` = sw$coef.of.var,
              `Scaled MAD`        = sw$scaled.mad,
              `Negative entropy`  = sw$negative.entropy,
              `Zero weights`      = sw$num.zeros)
 colnames(out) <- c("Treated", "Control")
 out
}

# i. Summary of the weights (range, dispersion, effective sample size)
sum.glm  <- summary(W.glm)
sum.ebal <- summary(W.ebal)

weights_table(sum.glm) |>
 kbl_pdf(caption = "Summary of weights: propensity score weighting (glm)")
```

Table 8: Summary of weights: propensity score weighting (glm)

	Treated	Control
Min. weight	1	0.000
Max. weight	1	73.222
Coef. of variation	0	16.002
Scaled MAD	0	1.778
Negative entropy	0	4.369
Zero weights	0	71.000

```
sum.glm$effective.sample.size |>
  kbl_pdf(caption = "Effective sample size: propensity score weighting (glm)", digits = 1)
```

Table 9: Effective sample size: propensity score weighting (glm)

	Control	Treated
Unweighted	2,490.0	185
Weighted	9.7	185

```
weights_table(sum.ebal) |>
  kbl_pdf(caption = "Summary of weights: entropy balancing")
```

Table 10: Summary of weights: entropy balancing

	Treated	Control
Min. weight	1	0.000
Max. weight	1	274.588
Coef. of variation	0	10.274
Scaled MAD	0	1.680
Negative entropy	0	3.488
Zero weights	0	14.000

```
sum.ebal$effective.sample.size |>
  kbl_pdf(caption = "Effective sample size: entropy balancing", digits = 1)
```

```
# ii. Covariate balance tables
bal.glm  <- bal.tab(W.glm,  stats = c("m", "v"), thresholds = c(m = .1), un = TRUE)
bal.ebal <- bal.tab(W.ebal, stats = c("m", "v"), thresholds = c(m = .1), un = TRUE)

bal.glm$Balance |>
  kbl_pdf(caption = "Covariate balance: propensity score weighting (glm)", wide = TRUE)
```

Table 11: Effective sample size: entropy balancing

	Control	Treated
Unweighted	2,490.0	185
Weighted	23.4	185

Table 12: Covariate balance: propensity score weighting (glm)

	Type	Diff.Un	V.Ratio.Un	Diff.Adj	M.Threshold	V.Ratio.Adj
prop.score	Distance	2.215	11.012	-0.341		0.897
age	Contin.	-1.263	0.470	0.280	Not Balanced, >0.1	1.231
education	Contin.	-0.881	0.425	0.030	Balanced, <0.1	0.644
black	Binary	0.593		0.025	Balanced, <0.1	
hispanic	Binary	0.027		-0.061	Balanced, <0.1	
married	Binary	-0.677		0.096	Balanced, <0.1	
nodegree	Binary	0.403		-0.008	Balanced, <0.1	
re74	Contin.	-3.547	0.133	0.135	Not Balanced, >0.1	1.302
re75	Contin.	-5.446	0.056	0.058	Balanced, <0.1	1.352
u74	Binary	0.622		-0.104	Not Balanced, >0.1	
u75	Binary	0.500		0.187	Not Balanced, >0.1	

```
bal.glm$Observations |>
  kbl_pdf(caption = "Effective sample sizes: propensity score weighting (glm)", digits = 1)
```

Table 13: Effective sample sizes: propensity score weighting (glm)

	Control	Treated
Unadjusted	2,490.0	185
Adjusted	9.7	185

```
bal.ebal$Balance |>
  kbl_pdf(caption = "Covariate balance: entropy balancing", wide = TRUE)
```

```
bal.ebal$Observations |>
  kbl_pdf(caption = "Effective sample sizes: entropy balancing", digits = 1)
```

```
# iii. Love plots of standardized mean differences
love.plot(W.glm,  stats = "mean.diffs", abs = TRUE, binary = "std",
        thresholds = c(m = .1), var.order = "unadjusted",
        title = "Covariate balance: propensity score weighting (glm)")
```

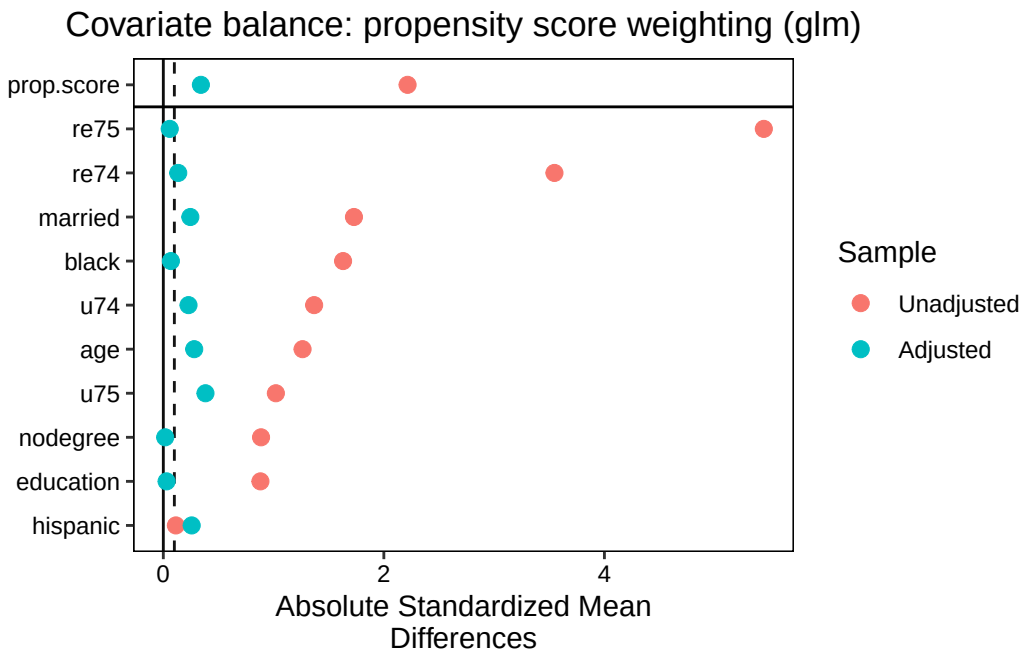
Table 14: Covariate balance: entropy balancing

	Type	Diff.Un	V.Ratio.Un	Diff.Adj	M.Threshold	V.Ratio.Adj
age	Contin.	-1.263	0.470	0	Balanced, <0.1	0.755
education	Contin.	-0.881	0.425	0	Balanced, <0.1	0.675
black	Binary	0.593		0	Balanced, <0.1	
hispanic	Binary	0.027		0	Balanced, <0.1	
married	Binary	-0.677		0	Balanced, <0.1	
nodegree	Binary	0.403		0	Balanced, <0.1	
re74	Contin.	-3.547	0.133	0	Balanced, <0.1	1.081
re75	Contin.	-5.446	0.056	0	Balanced, <0.1	1.012
u74	Binary	0.622		0	Balanced, <0.1	
u75	Binary	0.500		0	Balanced, <0.1	

Table 15: Effective sample sizes: entropy balancing

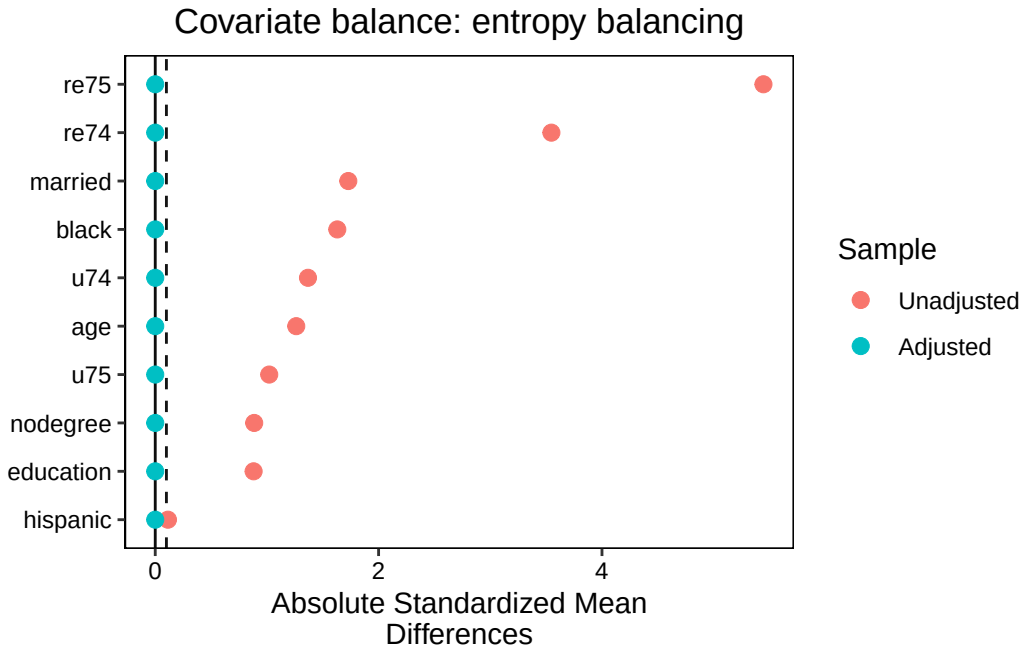
	Control	Treated
Unadjusted	2,490.0	185
Adjusted	23.4	185

Warning: Large mean differences detected; you may not be using standardized mean differences for continuous variables.



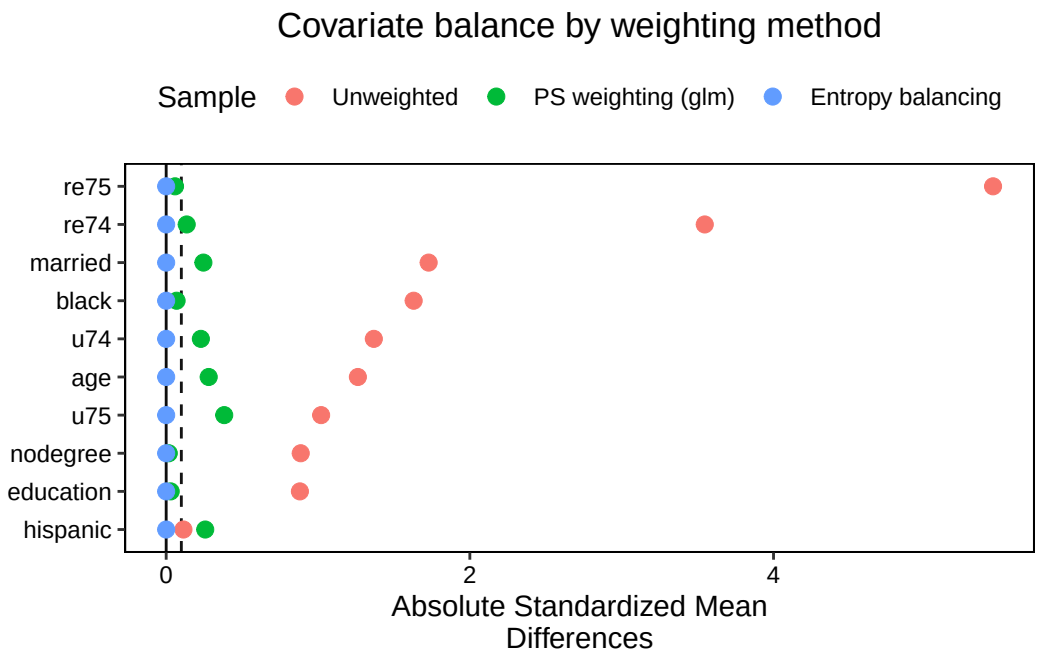

```
love.plot(W.ebal, stats = "mean.diffs", abs = TRUE, binary = "std",
        thresholds = c(m = .1), var.order = "unadjusted",
        title = "Covariate balance: entropy balancing")
```

Warning: Large mean differences detected; you may not be using standardized mean differences for continuous variables.




```
# Both methods in the same plot
love.plot(f_w, data = lalonde.psid, estimand = "ATT",
        weights = list(glm = W.glm, ebal = W.ebal),
        stats = "mean.diffs", abs = TRUE, binary = "std",
        thresholds = c(m = .1), var.order = "unadjusted",
        sample.names = c("Unweighted", "PS weighting (glm)", "Entropy balancing"),
        title = "Covariate balance by weighting method") +
 theme(legend.position = "top")
```

Warning: Large mean differences detected; you may not be using standardized mean differences for continuous variables.



(d) Estimate the treatment effect, and interpret your results.

```
# Outcome model formula: treatment interacted with the raw covariates
f_out <- as.formula(paste0("re78 ~ treat * (", paste(covs_raw, collapse = " + "), ")"))

# Prior earnings (re74, re75) are in dollars, so the covariates in the
# outcome model have very different scales. Combined with the interactions
# and the highly concentrated weights, this makes the model's Hessian
# numerically near-singular, and glm_weightit() can fail when computing
# standard errors. Measuring prior earnings in thousands of dollars fixes
# this without changing the estimated effect (a linear model is invariant
# to rescaling its covariates). The weights are unaffected because the rows
# are the same as in the data used by weightit().
lalonde.psid.k <- lalonde.psid |>
  mutate(re74 = re74 / 1000,
         re75 = re75 / 1000)

## Propensity score weighting (glm)
# Weighted difference in means (robust SEs that account for weight estimation)
fit_glm_dim <- glm_weightit(re78 ~ treat, data = lalonde.psid.k, weightit = W.glm)
coef_table(fit_glm_dim) |>
  kbl_pdf(caption = "PS weighting (glm): difference in means")
```

```
# Weighted regression adjustment, averaged over the treated (ATT)
fit_glm <- glm_weightit(f_out, data = lalonde.psid.k, weightit = W.glm)
avg_comparisons(fit_glm,
                variables = "treat",
```

Table 16: PS weighting (glm): difference in means

Term	Estimate	Std. Error	z value	p-value
(Intercept)	3,553	553.3	6.421	<0.001
treat	2,796	778.7	3.591	<0.001

```
            newdata   = subset(lalonde.psid.k, treat == 1)) |>
 ate_table() |>
 kbl_pdf(caption = "PS weighting (glm): regression-adjusted ATT", digits = 2)
```

Table 17: PS weighting (glm): regression-adjusted ATT

Term	Contrast	Estimate	Std. Error	z	p-value	CI lower	CI upper
treat	1 - 0	2,413.86	648.3	3.72	<0.001	1,143.21	3,684.51

```
## Entropy balancing
# Weighted difference in means
fit_ebal_dim <- glm_weightit(re78 ~ treat, data = lalonde.psid.k, weightit = W.ebal)
coef_table(fit_ebal_dim) |>
 kbl_pdf(caption = "Entropy balancing: difference in means")
```

Table 18: Entropy balancing: difference in means

Term	Estimate	Std. Error	z value	p-value
(Intercept)	3,924	230.6	17.02	<0.001
treat	2,425	621.2	3.903	<0.001

```
# Weighted regression adjustment, averaged over the treated (ATT)
fit_ebal <- glm_weightit(f_out, data = lalonde.psid.k, weightit = W.ebal)
avg_comparisons(fit_ebal,
            variables = "treat",
            newdata   = subset(lalonde.psid.k, treat == 1)) |>
 ate_table() |>
 kbl_pdf(caption = "Entropy balancing: regression-adjusted ATT", digits = 2)
```

Table 19: Entropy balancing: regression-adjusted ATT

Term	Contrast	Estimate	Std. Error	z	p-value	CI lower	CI upper
treat	1 - 0	2,424.66	646.78	3.75	<0.001	1,157.01	3,692.32

3.1 Extensions

(e) Repeat b) to d) using an alternative weighting method.

(f) How do results compare with the previous? Assuming you *cannot* see the experimental benchmark, what estimate would you prefer and why?